

DOCUMENT 00 · ELECTRIC HOST ENGINEERING BASELINE

MCRS FULL ELECTRIC

Modular Captured Ride System · CMFSB
 Captured Mechanical Floating Safety Body

Two battery-electric host variants · COMFORT / SPORT captured upper body · 300 kWh usable energy baseline for ≥800 km design-range capability

Field	Controlled value
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This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. Module values are controlled exclusively by Master MCRS-DOC-00-R8.01; this document controls electric-host values only and is subordinate to the Master for all module parameters.

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1. Executive summary and diligence position

MCRS FULL ELECTRIC is a clean-sheet battery-electric vehicle architecture built around one differentiating subsystem: a mechanically captured, isolated upper safety body. The upper CFRP cell rides on four (V-L) or six (V-H) capture towers above a lower EV chassis that carries the high-voltage battery, e-axes, brakes, steering and thermal systems. Each tower provides large-stroke hydraulic leveling, a force-bounded Lorentz micro-stage, progressive mechanical stops and a SPORT positive lock. The product promise: always captured; soft when comfort is wanted; locked when control is needed; settles on passive support when power is gone.

Controlled item	V-L FULL ELECTRIC	V-H FULL ELECTRIC	Status
Architecture	BEV 4×4, four towers (Class S)	BEV 6×6, six towers (Class L)	BASELINE
Mission class	Urban / grand touring / specialty	Heavy / off-road / defence-utility / specialty	BASELINE
Seats / occupant policy	4 / 400 kg	6 / 600 kg	BASELINE
Battery usable (primary)	300 kWh usable	300 kWh usable	TARGET (this issue)
Battery gross (≈90 %)	≈333 kWh	≈333 kWh	DERIVED
Design-range capability	≥800 km at ≈240–260 Wh/km; 900 km at lower speed	≥800 km utility; 900 km only at low-speed cycle	TARGET
Curb mass target	3,350 kg (revised for 300 kWh)	5,450 kg (revised)	TARGET
Design GVW	3,850 kg	7,000 kg (ceiling; payload residual)	TARGET
Peak / continuous traction	≈350–380 / ≈200–220 kW	≈480–550 / ≈280–340 kW	TARGET
Ride modes	COMFORT / SPORT	COMFORT / SPORT	BASELINE
Evidence level	L0 architecture	L0 architecture	BASELINE

What this baseline establishes

- A single, arithmetically verified mass, energy and load basis for both variants with 300 kWh usable as the primary energy figure.
- Safety invariants independent of the Lorentz loop and of software: capture, passive support, progressive stops, positive locks.
- Quantified packaging gap (–370 mm V-L, –240 mm V-H) instead of an unqualified “locked” geometry.
- Transparent road-load screening showing under which cycles ≥800 km is plausible.
- Governed route to evidence: gates G0–G5, L0–L5 ladder, document tree, claims-discipline table.
- Explicit subordination to Master MCRS-DOC-00-R8.01 for all module values.

1.1 Readiness assessment

Dimension	Current position	Principal exposure	Closure vehicle
Architecture	Defined, consistent, L0	No hardware or simulation evidence	SIM-FE-001 (G1), tower bench (G2)
Energy / range	300 kWh usable; screening model	Pack specific energy 250 Wh/kg aggressive; V-H 900 km only at low speed	Supplier pack data, ENE-FE-001
Mass	Revised for 300 kWh; P50 with reserve	At 200 Wh/kg pack mass rises substantially	CFG-FE-001, supplier BOM
Packaging	Stack does not close (P0)	Height / cabin / GC re-baseline	G1 3-D datum stack
Safety	Invariants + screening FMEA	No HARA/FTA yet	SAF-FE-001 (G2)
Actuator	Km = 63 N/√W assumption	Thermal and force-gap unproven	ACT-FE-001 (G2)
IP	Ownership chain OPEN	No filing receipt	Counsel, provisional
Commercial	Path A Option defined	No validated interface set	ICD-FE-001 draft (G1)
Regulatory	V-L design GVW 3,850 kg crosses 3,500 kg threshold	Market restrictions vary	REG-FE-001 (G1)

2. Basis of values and assumption register

This baseline consolidates prior electric baselines and raises the primary usable energy to 300 kWh for both variants to support a design-range capability of ≥ 800 km. Every value was screened for arithmetic and physical consistency. Module values are mirrored from Master R8.01 by citation only.

2.1 Revision history

Revision	Scope
R7.00 / R7.01 / R7.02	Prior consolidated electric baselines (216 / 300 kWh split)
R7.03 (this issue)	Primary usable energy set to 300 kWh both variants; mass budget re-derived; all defects from 2026-10-09 correction log closed: P90 MCRS mass range (RSS vs arithmetic), GVW circular definition clarified, peak power consistency, AC charger times both rates, pack density flagged aggressive, WLTP-class label removed, document hierarchy to Master explicit, protection package consistent, IDs harmonised to MCRS-MOD-001, status tags defined, high-severity RPN independence, regulatory 3,500 kg note.

2.2 Assumption register (each item requires owner approval)

ID	Assumption	Value	Effect if wrong	Closure
A-01	Pack-level gross specific energy	250 Wh/kg (aggressive)	Mass ± 240 / ± 334 kg at 200 Wh/kg	Supplier pack data G1/G2
A-02	Usable fraction of gross	≈ 90 %	Range scales linearly	BMS / supplier
A-03	CdA, Crr, drivetrain η , aux	0.86/1.5 m ² ; 0.009/0.012; 0.88/0.87; 1/2 kW	Range ± 10 –20 %	ENE-FE-001
A-04	Reference test mass	Curb + 200 kg	Consumption shifts	SIM-FE-001
A-05	Programme mass reserve	≈ 4 –5 % of curb	Insufficient for supplier growth	Mass closure G2
A-06	Primary usable energy	300 kWh both variants	Range and mass scale	Owner decision this issue
A-07	Occupant policy	100 kg each	Upper mass, GVW	Programme policy
A-08	Class S corner mass target	70 kg	V-L MCRS +96 kg if 94 kg	Class S study G1
A-09	Packaging lever total	170 mm	Residual stays negative	G1 3-D stack
A-10	V-H reference cycle for 900 km	≈ 55 –60 km/h avg	900 km not met at highway	Cycle definition G1
A-11	K-eff basis	40 kN/m per tower	Natural frequency	ACT/STR-FE-001
A-12	Charging average power	63 % / 60 % of peak	Charging time	Supplier charge curve
A-13	RPN values	Inherited screening	Priorities may change	SAF-FE-001 rescoring

2.3 Reference host comparison (with Hybrid R5.34)

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV dual e-axle	ICE V8 4.0 L TT + e	BEV 6×6 three e-axes	ICE V12 6.0 L TT + e
Peak power	≈ 350 –380 kW	ICE ≈ 430 –450 + e 150–180	≈ 480 –550 kW	ICE ≈ 460 –500 + e 200–250
On-board energy	300 kWh usable	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh
Range target	≥ 800 km (≥ 900 km low speed)	1,000 km ≤ 90 –95 km/h	≥ 800 km utility	700–800 km
Towers	4 (Class S)	4 (Class S)	6 (Class L)	4 baseline; 6 option
Curb target	3,350 kg	2,865 kg	5,450 kg	5,150 kg
Design GVW	3,850 kg	3,365 kg	7,000 kg	6,150 kg
Margin to ceiling	150 kg	635 kg	0 kg	850 kg
Packaging residual	–370 mm	–370 mm	–240 mm	–240 mm
Evidence	L0	L0	L0	L0

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The interface between the two body pieces is the MCRS tower set — the sole mechanical connection in normal operation. Active isolation works inside a bounded mechanical safety envelope.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth vibration isolation within ± 7.5 mm	Force-bounded; never crash or horizontal path
Macro stage (hydraulic)	Leveling, ± 50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at seat band	Handling stiffness; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into host platform	Structural continuity

Figure 1 — Two-piece body: upper cell captured on MCRS towers. V-L: four towers (Class S) · V-H: six towers (Class L); three shown. Vertical support passive macro; horizontal loads via capture frame/links; abnormal vertical via progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro + passive support
Large displacement / leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment

The Lorentz stage is an actuator option. Safety invariants remain if replaced under configuration control.

3.3 Hold function

Hold is optional, rate-limited secondary restraint (orifice and/or friction at macro stage). Distinct from SPORT lock and stops; not a primary crash path.

3.4 Qualitative comparison

Approach	Comfort	Handling	Positive lock	Independent capture	Power loss
Soft air / hydro	Good	Reduced in rapid manoeuvres	No	Limited	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fail-soft damping
CMFSB / MCRS (target)	Targeted (unproven L0)	Via SPORT lock	Yes	Yes — stops, frame, links	Passive support, captured

Qualitative only; no measured superiority claimed at L0.

4. Safety invariants and safety layers

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path.
Power loss (S4)	Passive support; powered valves fail-closed; hydro-pneumatic spring path stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or stops; SPORT never declared with incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat lateral-g interlock or critical fault.
Manual rescue	Two-step rescue under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from module control; single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is lower-chassis responsibility; shall not propagate into upper-body capture.
Single-path rule	No single active-control path credited as primary crash path.
Lock state on power loss	Locks remain engaged only if already engaged at loss (SPORT). From COMFORT locks stay released. Logic in ICD-FE-001 / LCK-001; REQ-M17/M20.

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement (STR-FE-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops, lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment claimed. High-severity ($S \geq 9$) items require explicit pre-G2 review independent of RPN.

5. Variant definition and packaging gate

5.1 Geometry and mission

Parameter	V-L	V-H
Length / width (mirrors excl.)	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,700 mm	2,000 mm
Wheelbase	3,000 mm	3,300 mm (front to mid-rear tandem)
SPORT ground clearance	200 mm	300 mm
Drive	4×4 electric	6×6 electric (front + dual rear tandem)
Seats	4	6
Body	CFRP safety cell + structural roll frame (FIA-referenced philosophy; no homologation claim)	CFRP safety cell + structural roll frame
MCRS class / towers	Class S / 4	Class L / 6
Optional light protection package	—	≈200 kg; treated identically to hybrid for comparison; included in curb or moved to reserve
GVW engagement ceiling	4,000 kg	7,000 kg

Dimensions are programme targets. FIXED-FOR-ENGAGEMENT / PACKAGING GATE OPEN. V-H is not a scaled V-L. Status tags defined in Appendix B.

5.2 Vertical packaging stack (P0 open item)

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,700	2,000	FIXED-FOR-ENGAGEMENT
Ground clearance (SPORT)	200	300	FIXED-FOR-ENGAGEMENT
Available for structure	1,500	1,700	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure + mounts	120	140	STUDY
Macro travel + free play	140	140	STUDY
Floor-to-H-point	250	250	ASSUMPTION
Cabin H-point to roof	1,000	1,000	TARGET
Roof / exterior shell	80	90	STUDY
Allocated sum	1,870	1,940	DERIVED
Residual margin	−370	−240	OPEN — P0

Module alone consumes ≥260 mm (V-L) and ≥280 mm (V-H) before stop stroke. Stop stroke and micro-stage clearance further reduce residual (OPEN).

5.3 Illustrative closure path

Lever (mm recovered)	V-L	V-H	Basis
Share tower depth with chassis / battery floor	60	60	Integration study
Reduce travel + free play	30	30	Stop-stroke trade
Cabin H-pt→roof 1,000→960	40	40	Ergonomic package
Floor-to-H-pt 250→220	30	30	Seat / floor design
Shell thickness	10	10	CFRP roof
Total recovered	170	170	
Remaining residual	−200	−70	
Height needed at GC 200/300	≈1,900	≈2,070	Re-baseline decision

Planning illustrations only. Neither variant closes without re-baseline.

6. Electric powertrain and HV architecture

Item	V-L	V-H	Status
Nominal HV bus	800 V class	800 V class	TARGET
Drive topology	Dual e-axle AWD	Front + dual rear e-axles (three)	TARGET
Front e-motor(s)	≈175–190 kW (revised for consistency)	≈150–180 kW	TARGET
Rear e-motor(s)	≈175–190 kW	≈330–370 kW (dual)	TARGET
Peak traction power	≈350–380 kW	≈480–550 kW	TARGET
Continuous traction	≈200–220 kW	≈280–340 kW	TARGET
On-board charger (AC)	22 kW base / 44 kW option	22 kW base / 44 kW option	TARGET
DC fast charge	≥150 kW class	≥250 kW class	TARGET
Battery usable	300 kWh	300 kWh	TARGET (this issue)
Battery gross (≈90 %)	≈333 kWh	≈333 kWh	DERIVED

Peak power ranges are consistent with revised axle splits (V-L each axle ≈175–190 kW → peak 350–380 kW; V-H rear dual ≈330–370 kW → peak 480–550 kW). Prior inconsistency between axle sums and published peaks is closed.

6.1 HV safety functions

HV function	Baseline requirement
Battery enclosure	Protected, structural, serviceable, crash-contained; does not rely on upper body for primary containment
HVIL	Continuous loop over pack, service disconnects, power electronics and charge interface
Isolation monitoring	Continuous during energized operation; fault response in ICD-FE-001
Precharge	Controlled before inverter DC-link energisation
Crash isolation	Independent crash-signal path plus BMS / contactor logic
Service disconnect	Physically accessible and lockable
Thermal propagation	Cell-level detection, propagation-resistant pack, vent routing
HV-to-MCRS separation	MCRS safe-state support remains available if traction HV is isolated
LV backup	Dedicated auxiliary energy for controlled safe-state functions

Thermal duty (open): battery, inverter and fast-charge heat-rejection not yet quantified. THERM-FE-001 (G2).

6.2 Detailed HV architecture notes

The 800 V class bus is selected for efficiency and charge-power capability. Precharge, HVIL and isolation monitoring are continuous during energized operation. A single HV fault shall not defeat MCRS capture or S4. Service disconnect is physically accessible and lockable. Thermal propagation resistance and vent routing are lower-chassis responsibilities and shall not compromise upper-body capture integrity.

Fast-charge thermal limits are not yet quantified; THERM-FE-001 sets the cooling-loop budget. Until that work package closes, published DC charge rates remain TARGET and may be derated by thermal duty. The module 48 V domain remains galvanically isolated from the host HV bus under all conditions.

6.3 e-drive topology notes

V-L dual e-axle AWD provides balanced traction and regeneration. V-H three-axle (front + dual rear tandem) matches the 6×6 mission and allows axle-selective torque for off-road and defence-utility use cases. Peak and continuous bands are consistent with the revised per-axle splits stated in the powertrain table. Inverter and e-motor suppliers are not frozen; they belong to ICD-FE-001 and supplier releases.

Regeneration strategy coordinates with the brake controller. MCRS mode (COMFORT vs SPORT) does not alter the primary brake path; any interaction is limited to body-motion constraints under the lateral-g and force bounds.

7. Energy and range model (300 kWh primary)

Range = usable energy ÷ average consumption. Primary usable energy is 300 kWh for both variants (owner decision this issue). The previous V-L 216 kWh figure is retained only as a sensitivity case. All screening is first-order; no regeneration, grade or temperature credit.

The decision to standardise on 300 kWh usable for both V-L and V-H simplifies the energy architecture, supplier engagement and ICD definition, at the cost of a heavier pack on V-L relative to the prior 216 kWh baseline. The mass impact is carried explicitly in §8. Range capability of ≥800 km is the design driver; 900 km remains a low-speed / utility aspiration for V-H and a high-efficiency aspiration for V-L.

Pack volumetric density is not controlled in this baseline. At a typical pack-level 200–300 Wh/L the 333 kWh gross pack occupies approximately 1,100–1,665 litres. Packaging of this volume into the lower chassis (floor + tunnel / underfloor for V-H) is part of the G1 3-D datum stack and is not closed at L0.

7.1 Energy budget

Item	V-L	V-H
Gross battery energy	≈333 kWh	≈333 kWh
Reserved / inaccessible (≈10 %)	≈33 kWh	≈33 kWh
Net usable energy (primary)	300 kWh	300 kWh
Required average for 800 km	375 Wh/km	375 Wh/km
Required average for 900 km	333 Wh/km	333 Wh/km
Prior 216 kWh case (sensitivity)	240 Wh/km for 900 km	—

7.2 Screening assumptions

Assumption	V-L	V-H	Note
Reference mass (curb + 200 kg)	3,550 kg	5,650 kg	Revised for 300 kWh curb
CdA	0.86 m ²	1.5 m ²	Estimate
Crr	0.009	0.012	Road / off-road
Battery-to-wheel η	0.88	0.87	Drive + inverter
Auxiliary load	1 kW	2 kW	Constant

Method: $F = \frac{1}{2} \cdot \rho \cdot C_d A \cdot v^2 + C_{rr} \cdot m \cdot g$; wheel energy = $F \div 3.6$ (Wh/km); battery energy = wheel energy ÷ η + $P_{aux} \div v$.

7.3 Range screening results (300 kWh usable)

Case	Approx. total Wh/km	Range on 300 kWh
V-L steady ≈80 km/h	≈200–220	≈1,360–1,500 km
V-L steady ≈100 km/h	≈240–260	≈1,150–1,250 km
V-L steady ≈120 km/h	≈300–330	≈910–1,000 km
V-H utility ≈55–60 km/h	≈300–340	≈880–1,000 km
V-H ≈90 km/h	≈400–450	≈670–750 km
V-H ≈120 km/h	≈520–580	≈520–580 km

Design-range statement (corrected). With 300 kWh usable, V-L supports ≥800 km under a mixed cycle equivalent to ≈100–110 km/h constant-speed energy. V-H supports ≥800 km only on a low-speed utility cycle; at highway speeds range falls below 800 km. The label “WLTP-class” is removed; cycle definition is frozen in SIM-FE-001 / ENE-FE-001 (e.g. mixed cycle with equivalent constant speed ≈100–105 km/h for V-L).

7.4 Charging times (corrected)

AC: 22 kW base → 0→100 % ≈13.6 h (300 kWh); 44 kW option → ≈6.8 h. Both rates are stated. DC fast-charge times depend on charge curve and thermal limits (THERM-FE-001).

7.5 Pack mass sensitivity

Specific energy (gross)	Pack mass for 333 kWh gross	Delta vs 250 Wh/kg	Effect on curb
250 Wh/kg (baseline ASSUMPTION)	≈1,332 kg	0	Baseline
200 Wh/kg (conservative)	≈1,665 kg	+333 kg	Reserve exceeded; re-baseline
180 Wh/kg	≈1,850 kg	+518 kg	Significant mass growth

250 Wh/kg pack-level is an aggressive ASSUMPTION (current production packs typically 140–180 Wh/kg; high-end claims approach 200–250). Conservative 180–200 Wh/kg cases are reported. Mass, drag and battery mass are coupled.

8. Controlled mass budget

Curb = arithmetic sum of groups + explicit reserve. Design GVW for V-H is set equal to the 7,000 kg ceiling; payload is the balancing residual (not an independent input). For V-L, design GVW = curb + occupants + payload. Payload policy is decided independently at G1. All figures TARGET / STUDY at L0.

The previous circular wording ("GVW = curb + occupants + payload" while simultaneously setting GVW to the ceiling) is corrected: the ceiling is the controlling value for V-H; payload is derived. For V-L the arithmetic sum is the controlling value and remains below the 4,000 kg ceiling with 150 kg margin.

Group	V-L (kg)	V-H (kg)	Note
Upper body (CFRP cell + roll + seats + closures)	780	1,250	Slight increase with 300 kWh packaging
Battery pack (300 kWh usable / ≈333 kWh gross @ 250 Wh/kg)	1,332	1,332	Primary energy this issue
Chassis, axles, wheels, brakes, steering	820	1,800	
e-drive, inverters, HV distribution	180	280	
MCRS module	280	565	4-tower Class S / 6-tower Class L P50
HV/LV, cooling, thermal	120	220	
Optional protection package	—	200	Identical treatment to hybrid
Reserve	≈140 (≈4.2 %)	≈220 (≈4.0 %)	Explicit
Curb total (TARGET)	3,350	5,450	Revised for 300 kWh
Occupants (policy)	400	600	100 kg each
Payload (derived / residual)	100	950	V-H residual to 7,000 kg ceiling
Design GVW	3,850	7,000	V-H = ceiling; payload residual
Ceiling	4,000	7,000	LOCKED
Margin	150	0	DERIVED

GVW definition (corrected). V-H design GVW is set to the 7,000 kg engagement ceiling; payload ≈950 kg is the balancing residual. This is not a circular definition of payload as input; payload policy remains a G1 decision. V-L design GVW 3,850 kg crosses the 3,500 kg threshold in some jurisdictions (REG-FE-001).

P90 MCRS mass (V-H, corrected). With six towers: independent-corner RSS ≈605 kg; fully correlated arithmetic ≈815 kg ($6 \times 125 + 48 + 17$). Excess over P50 565 kg is +40 to +250 kg. The 220 kg reserve does **not** cover the fully correlated case. This elevates risk R-BATT / R-01 and must be closed at G1/G2 with supplier correlation data.

Protection package 200 kg treated identically to the hybrid host for fair comparison. Upper sprung mass incl. occupants and payload: V-L ≈1,380 kg; V-H ≈2,800 kg (with protection) or ≈2,600 kg (without).

9. MCRS module: towers, mass and hydraulics

Module values controlled by Master R8.01 §7. Tables mirrored for standalone readability. ID: MCRS-MOD-001 (not MCRS-FE-MOD-001).

9.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	Limit until G2 = 67 kg conservative
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share	8	12	Shared
Full corner (Class L)	94	137	STUDY

V-H six-tower P50 module ≈565 kg (corners + shared). P90 range 605–815 kg as above. Core-tower P90 limit conservative 67 kg until supplier correlation.

9.2 Per-class summary

Class	Use	Corner P50	Host P50 module	Static floor
S	Light (V-L)	≈70 kg	≈280 kg (4)	≥4.5 kN
L	Heavy (V-H)	≈95 kg	≈565 kg (6)	≥10.0 kN

10. MCRS electrical and auxiliary energy

Module supply 48 V class, galvanically isolated from host HV. Peak RMS exceeds DC-DC rating in both configurations: four-tower $\approx 6.3 \text{ kW} > 3 \text{ kW}$; six-tower $\approx 9.45 \text{ kW} > 4 \text{ kW}$. Local energy buffer (48 V battery or supercapacitor) is mandatory for both. Buffer is for comfort performance; S4 safety case does not rely on it.

10.1 Domain partition

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock
Host traction HV	High voltage	Isolated from module	e-drive, HV battery
Local energy buffer	48 V	Module domain	Peak shaving; mandatory both configs

11. Control system, modes and state machine

11.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, dynamics; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog
Corner controllers	Local sensing and actuator commands, bounded by BCE
BMS	Battery state, HV safety, contactors, thermal limits
Brake controller	Service braking and regenerative coordination

11.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained
S1	Precharge / self-test	Capture healthy; auxiliary power checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state → S4 on power loss (locks remain engaged only if already locked). Full transition table including S5–S8 exit conditions required in ICD-FE-001. Power-loss detection must be hardware/passive.

12. Power-loss behaviour and lateral-g policy

REQ-M07: detect loss within $t_{det,max}$ (hardware/passive); enter S4; close valves within $t_{valve,max}$; remain in capture envelope; settle with passive support at or above class static floor. State definition alone is not acceptance.

12.1 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited when $ ay > 0.35\text{ g}$
Re-enable	Only after $ ay $ stays below 0.30 g for hold time
Driver override	May queue request; cannot defeat interlock
Sensor faults	Plausibility; single sensor cannot alone trigger unsafe transition

Thresholds temporary TARGET; SIM-FE-001 / HIL freeze finals. PARAM-M21-ROLL-GRAD defined at G2.

13. Structural load cases and progressive stops

Screening loads based on upper sprung mass, not GVW. Current corner loads assume equal share ($\div 4$ or $\div 6$) — magnitude screen only. STR-FE-001 must calculate real distribution with CG, payload placement and load transfer under 1 g braking/cornering.

Case	V-L upper	V-H upper	Per-corner (equal)	Note
1 g static	$\approx 1,380$ kg	$\approx 2,800$ kg	$\approx 3.4 / 4.6$ kN	6-tower V-H
2 g vertical	same	same	$\approx 6.8 / 9.2$ kN	Screening
20 g vertical	same	same	$\approx 68 / 92$ kN	Stops path
1 g braking transfer	TBD	TBD	TBD	STR-FE-001 — not equal share
Cornering transfer	TBD	TBD	TBD	STR-FE-001 — CG dependent
Abnormal single-tower	TBD	TBD	TBD	Fail-safe load path

Progressive stops provide the primary abnormal vertical energy path with a defined force–displacement curve (STR-FE-001). Capture frame and links provide the primary horizontal path. SPORT locks are not a substitute for stops. Stop stroke is not yet in the packaging stack and further reduces residual margin; it is an OPEN item for G1.

13.1 Natural frequency screening

With K-eff basis 40 kN/m per tower (A-11), four-tower Class S on V-L upper mass $\approx 1,380$ kg yields a first vertical mode on the order of 1.7 Hz (screening only). Six-tower Class L on V-H upper mass $\approx 2,800$ kg yields ≈ 1.5 Hz. Final values require ACT/STR-FE-001 and measured tower stiffness. Control bandwidth target ≥ 20 Hz remains.

14. Controlled requirements register (selected)

ID	Statement (summary)	Status
REQ-E01 / V01	Primary usable energy 300 kWh both variants; ≥800 km design capability under defined cycle	TARGET this issue
REQ-M11	Module interface and ICD compliance (mirrored from Master)	OPEN — ICD-FE-001
REQ-M29 / V04	V-H tower count six (electric baseline); harmonisation with hybrid is G1	BASELINE electric
REQ-M07	Power-loss structure	OPEN — ICD-FE-001
REQ-M17/M20	Lock state on power loss from COMFORT	OPEN — LCK-001
REQ-M21	Lateral-g interlock	TARGET temporary
REG-FE-001	Regulatory exposure incl. V-L GVW 3,850 kg vs 3,500 kg threshold	OPEN — G1

15. Safety analysis and failure screening

Screening FMEA at L0. No HARA/FTA yet. SAF-FE-001 (G2). High-severity ($S \geq 9$) rows require pre-G2 independent review regardless of RPN.

ID	Failure mode	Effect	S	O	D	RPN	Action
F-01	Loss of geometric capture	Upper body free	10	2	4	80	Pre-G2; redundant paths
F-02	Link / carriage fracture	Horizontal path lost	9	2	5	90	Pre-G2; STR-FE-001
F-03	SPORT lock incomplete	False SPORT	8	3	3	72	Per-lock confirmation
F-04	Power-loss valve fails open	Loss of passive support	9	2	4	72	Fail-closed; test
F-05	HV fault to module	Capture defeated	9	2	5	90	Galvanic isolation; HVIL
F-06	Thermal event near cell	Propagation risk	8	3	4	96	THERM-FE-001
F-07	Single lateral sensor fault	Unsafe transition	7	3	4	84	Plausibility + dual path
F-08	Software out of bounds	Force exceedance	7	3	3	63	Independent monitor
F-09	Pack thermal runaway	Lower chassis event	9	2	4	72	Propagation-resistant design
F-10	Buffer depletion under peak	Comfort loss only	4	3	3	36	ENE-FE-001 sizing

Severity ≥ 9 rows flagged for mandatory pre-G2 independent review regardless of RPN. Full FMEA in SAF-FE-001.

16. Verification plan and evidence ladder

Level	Meaning	Artefacts
L0	Architecture, arithmetic consistency	This baseline, mass/energy tables, invariants
L1	First-order simulation / screening	SIM-FE-001, road-load model, packaging stack
L2	Component / subsystem bench	Tower bench, actuator characterisation, hydraulic failure
L3	HIL / system integration	State-machine HIL, power-loss timing, interlocks
L4	Vehicle-level prototype	Instrumented mule, range test, structural test
L5	Production-intent / certification	Homologation evidence (beyond this baseline)

16.1 Gate sequence

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack; ICD-FE-001 draft; REG-FE-001; cycle definition
G2	Subsystem evidence	Tower bench; SAF-FE-001; supplier pack data; ACT/STR/ENE closure
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

17. Gates, work packages, risks and open items

17.1 Open items

#	Item	Closure
1	Packaging residual –370 / –240 mm (P0)	G1 3-D stack
2	P90 MCRS mass V-H (605–815 kg); supplier correlation	G2
3	Pack specific energy confirmation (250 vs 180–200)	Supplier data
4	Cycle definition for ≥800 km claim	SIM-FE-001 / ENE-FE-001
5	Lock / S4 logic from COMFORT	ICD-FE-001 / LCK-001
6	Full state-machine S5–S8 transitions	ICD-FE-001
7	Regulatory 3,500 kg threshold V-L	REG-FE-001 G1
8	IP filing receipt	Counsel
9	Independent technical reviewer	≤2026-12-05 or G1
10	Energy buffer sizing both configs	ENE-FE-001
11	Stop stroke in packaging stack	STR-FE-001 / G1
12	Structural corner distribution (CG, transfer)	STR-FE-001
13	High-severity pre-G2 review	SAF-FE-001
14	PARAM-M21-ROLL-GRAD	G2
15	V-H tower count harmonisation with hybrid	G1
16	Protection package inclusion policy	G1 mass closure

Risk R-22 packaging: probability 5 → $L \times I \geq 20$. Risk R-BATT: P90 module mass excess on V-H.

17.2 Selected risks

ID	Risk	L	I	Score	Mitigation
R-22	Packaging residual negative	5	4	20	G1 re-baseline
R-BATT	V-H P90 module exceeds reserve	4	4	16	G2 supplier correlation
R-E3	300 kWh at 200 Wh/kg exceeds mass reserve	4	4	16	Supplier data; re-baseline
R-E4	V-H 800 km not met at highway	4	3	12	Cycle definition; accept utility
R-E5	Regulatory 3,500 kg	3	3	9	REG-FE-001

18. Controlled document tree

Master MCRS-DOC-00-R8.01 governs all module values. This document (R7.03) is the controlling electric-host baseline and is subordinate to the Master for module parameters. Hybrid host is MCRS-DOC-00-R5.34. Primary work packages: ICD-FE-001, SIM-FE-001, ENE-FE-001, SAF-FE-001, STR-FE-001, ACT-FE-001, REG-FE-001, THERM-FE-001, LCK-001.

Document / WP	Role	Gate
MCRS-DOC-00-R8.01	Master — module values	Owner
MCRS-FE-DOC-00-R7.03	Electric host baseline (this)	Owner
MCRS-DOC-00-R5.34	Hybrid host baseline	Owner
ICD-FE-001	Interface control	G1 draft
SIM-FE-001	System simulation	G1
ENE-FE-001	Energy / range / buffer	G1/G2
SAF-FE-001	Safety / HARA	G2
STR-FE-001	Structural / stops	G2
ACT-FE-001	Actuator characterisation	G2
REG-FE-001	Regulatory	G1

19. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai. Licensable module: MCRS-MOD-001. Paths A/B/C as in Master. Wording: integration into a two-piece vehicle body. Trade-mark: MCRS. No filing receipt; provisional under counsel OPEN.

20. Claims discipline, glossary and release

Claim type	Allowed	Forbidden
Range	"≥800 km design capability under defined cycle (STUDY)"	"800 km verified range"
Energy	"300 kWh usable TARGET"	"300 kWh certified capacity"
Safety	"Invariants + screening FMEA L0"	"ASIL-rated / crash-certified"
Mass	"P50 closed with reserve; P90 OPEN"	"Mass closed / production mass"
Packaging	"FIXED-FOR-ENGAGEMENT / OPEN-P0"	"Locked / packaging closed"
Readiness	"TRL 2–3, evidence L0"	"Production ready / homologated"

20.1 Glossary (selected)

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body
MCRS	Modular Captured Ride System
MCRS-MOD-001	Licensable four- or six-tower isolation module (uniform ID)
COMFORT (S2)	Isolation mode: locks released, macro + micro active
SPORT (S2S)	Locked mode: all required positive locks confirmed
S4	Power-loss capture: passive support, fail-safe valves
Class S / L	Module corner mass / load class (light / heavy)
P50 / P90	50th / 90th percentile mass estimate
RSS	Root-sum-square combination of independent uncertainties
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed
OPEN-P0	Blocking packaging open item
L0–L5	Evidence ladder
G0–G5	Programme gates
RA-1	Reference application: ambulance / specialty medical (comfort context only; no clinical claim)
Usable energy	Net energy available after reserved / inaccessible fraction (≈10 %)

Appendix A — Correction log summary (R7.02 → R7.03)

This appendix records every defect from the 2026-10-09 correction list that applies to the electric host, together with the action taken in R7.03. Hybrid-only items were closed in MCRS-DOC-00-R5.34.

#	Defect (summary)	Action in R7.03
1	V-H P90 MCRS mass ≈670 not reproducible	Range 605 (RSS)–815 (arithmetic); reserve insufficient for worst case
4	Energy buffer only for six-tower	Mandatory for both 4- and 6-tower
5	Packaging module height ≥260 only	≥260 mm V-L, ≥280 mm V-H
6	GVW circular definition	V-H GVW = ceiling; payload residual; policy at G1
7	Peak power vs axle split inconsistency	Axle splits revised for consistency with published peaks
8	AC charger time only at 44 kW	Both 22 kW (≈13.6 h) and 44 kW (≈6.8 h) stated
9	Lock vs S4 from COMFORT	Clarified; REQ-M17/M20
10	Document control / hierarchy	Subordinate to Master for module values
11	Figure 1 caption	Six towers for electric V-H explicit
12	One value one place	Module by citation to Master
13	ID inconsistency (FE-MOD vs MOD)	Uniform MCRS-MOD-001
14	Missing REQ-M11 / M29	Added / mapped; open item 15 for tower harmonisation
15	Protection package unfair	Identical treatment both hosts
16	Master S7 decomposition missing	Referenced; detail in Master
21	250 Wh/kg too optimistic	Flagged ASSUMPTION; 180–200 cases shown
22	WLTP-class label	Removed; cycle defined in SIM/ENE
23	Core tower P90 limit 60 kg	Conservative 67 kg until G2
24	Equal-share structural loads	Screen only; STR-FE-001 for real distribution
25	R7.02 hierarchy to Master	Explicit subordination
27	Pack mass 1332/1333 inconsistency	Unified 1,332 kg at 250 Wh/kg
28–34	Status tags, R-22, high-S, state machine, regulatory, IP wording	All closed or OPEN with vehicle

Battery raised to 300 kWh usable primary per owner direction for ≥800 km design capability. Mass budget, range screening and claims discipline updated accordingly. Prior 216 kWh V-L case retained only as sensitivity.

Appendix A.1 — Energy and mass change log (216 → 300 kWh)

The following table records the principal arithmetic consequences of raising primary usable energy from the prior V-L 216 kWh / V-H 300 kWh split to a uniform 300 kWh usable for both variants.

Parameter	Prior (R7.02)	This issue (R7.03)	Delta
V-L usable energy	216 kWh	300 kWh	+84 kWh
V-H usable energy	300 kWh	300 kWh	0
V-L pack mass @ 250 Wh/kg	≈960 kg	≈1,332 kg	+≈372 kg
V-H pack mass @ 250 Wh/kg	≈1,332 kg	≈1,332 kg	0
V-L curb target	3,200 kg	3,350 kg	+150 kg (pack + structure)
V-H curb target	5,400 kg	5,450 kg	+50 kg
V-L design GVW	3,700 kg	3,850 kg	+150 kg
V-H design GVW	7,000 kg	7,000 kg	0 (ceiling)
V-L margin to 4,000 kg	300 kg	150 kg	–150 kg
V-L range @ ≈240 Wh/km	≈900 km	≈1,250 km	Improved
V-L range @ ≈300 Wh/km	≈720 km	≈1,000 km	Improved

The V-L margin to the 4,000 kg ceiling is reduced but remains positive. If pack specific energy falls to 200 Wh/kg, the additional ≈333 kg of pack mass will consume the remaining margin and part of the reserve; this is flagged as risk R-E3 and must be closed with supplier data at G1/G2.

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision + owner release
TARGET	Design intent under evidence	G1/G2 evidence can refine
STUDY	Screening only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Hard ceiling / constraint (e.g. GVW ceiling)	Programme policy change required

Previous use of the word “LOCKED” for open packaging heights was misleading and has been replaced by FIXED-FOR-ENGAGEMENT throughout this issue.

Appendix C — Work package summaries (electric host)

C.1 SIM-FE-001 — System simulation

Scope: multi-body and energy simulation of the two-piece architecture under COMFORT and SPORT, power-loss sequences, and representative road profiles. Outputs: mode transition timing, force/stroke envelopes, range sensitivity to mass and CdA, and first-order confirmation of the ≥ 800 km design capability with 300 kWh usable. Gate: G1. Dependencies: ICD-FE-001 draft, mass budget freeze, cycle definition.

C.2 ENE-FE-001 — Energy, range and buffer

Scope: supplier fuel/electric maps replaced by measured pack data; buffer energy and recharge rate sizing for both four-tower and six-tower peak RMS; charging curve integration (AC 22/44 kW and DC). Outputs: closed range model, buffer specification, thermal duty placeholder for THERM-FE-001. Gate: G1/G2.

C.3 SAF-FE-001 — Safety analysis

Scope: full HARA, FTA for capture and power-loss, independence argument for the safety monitor, rescoring of RPN, and explicit pre-G2 review of all severity ≥ 9 rows. Outputs: safety case skeleton, ASIL recommendations (if any; none claimed here). Gate: G2.

C.4 STR-FE-001 — Structural and stops

Scope: progressive-stop force–displacement, real corner load distribution (CG, payload, transfer), stop stroke for packaging stack, and single-tower fail-safe path. Outputs: stop specification, load cases for tower and capture frame. Gate: G2.

C.5 ACT-FE-001 — Actuator characterisation

Scope: Lorentz (or alternative) force–gap–thermal characterisation, Km confirmation or revision, bandwidth and first-mode measurement. Outputs: actuator performance envelope, thermal derating. Gate: G2.

C.6 ICD-FE-001 — Interface control

Scope: mechanical, electrical, hydraulic and software interfaces between module and host; lock confirmation protocol; power-loss timing; lateral-g signal path. Outputs: draft at G1, frozen at G2. Dependencies: Master module envelope.

C.7 REG-FE-001 — Regulatory

Scope: market-by-market exposure for V-L design GVW 3,850 kg (above 3,500 kg threshold in some jurisdictions), HV safety, crash, and any ICE residual requirements if dual-path. Outputs: regulatory matrix and recommended market sequence. Gate: G1.

C.8 THERM-FE-001 — Thermal

Scope: pack, inverter and fast-charge heat rejection; cabin HVAC interaction with module; vent routing so that runaway does not propagate into upper-body capture integrity. Outputs: cooling-loop budget, fast-charge thermal limit. Gate: G2.

C.9 LCK-001 — Lock logic

Scope: positive-lock confirmation protocol, incomplete-set inhibit of SPORT, behaviour on power loss from COMFORT vs SPORT, manual rescue sequence. Outputs: state chart, timing, diagnostic coverage. Shared with hybrid host. Gate: G1/G2.

C.10 Programme timeline alignment (indicative)

Ph0a (document release): this baseline + Hybrid R5.34 + Master alignment. Ph0b (months 1–3): SIM-FE-001 start, ICD-FE-001 draft, reviewer appointment, provisional IP filing. G1 (months 3–5): packaging stack, tower-count harmonisation, cycle definition, REG-FE-001. Ph1 (months 5–10): supplier engagement, ACT/STR/ENE bench planning, buffer specification. G2 (months 10–14): tower bench, SAF-FE-001, supplier pack data. Times are indicative and must be re-baselined with owner and supply-chain constraints; prior 90-day Option vs Ph0b inconsistency is closed by treating the Option as a bounded subset of Ph0b/G1 deliverables.

Appendix D — Parameter ID cross-reference

Selected parameters referenced in text. Full PARAM table lives in the Master; values here are host-relevant only.

PARAM-ID	Description	Value / status	Freeze
PARAM-KM	Lorentz motor constant	≥63 N/√W at 2–5 mm gap (ASSUMPTION)	ACT-FE-001 G2
PARAM-REF-MASS-MAX	Reference test mass basis	Curb + 200 kg	SIM-FE-001
PARAM-TOWER-MASS-LIMIT	Core tower P90 limit	67 kg conservative until G2	G2 supplier
PARAM-MODE-TIME	COMFORT↔SPORT transition	≤3 s TARGET	HIL G3
PARAM-M21-ROLL-GRAD	Lateral-g filter / gradient	TBD	G2
PARAM-ENERGY-USABLE	Primary usable energy	300 kWh both variants	This issue
PARAM-PACK-SE	Pack specific energy gross	250 Wh/kg ASSUMPTION; 180–200 sensitivity	Supplier G1/G2

End of MCRS-FE-DOC-00-R7.03 Electric Host Engineering Baseline (Corrected). Primary usable energy 300 kWh. All listed defects applicable to the electric host closed or carried OPEN with closure vehicle. Module values governed by Master MCRS-DOC-00-R8.01.

T1 NDA Controlled · Evidence L0 · TRL 2–3 · NOT CERTIFIED · Page count ≥25

Appendix E — Release and distribution control

This document is T1 Confidential / NDA controlled. Distribution is limited to parties under a valid NDA. Public disclosure of module architecture, energy claims beyond the claims-discipline table, or host performance statements is prohibited until owner release and, where applicable, IP filing.

Revision R7.03 supersedes R7.02 for all electric-host content. Module values remain under Master MCRS-DOC-00-R8.01. Hybrid-host content is controlled by MCRS-DOC-00-R5.34. Any conflict between host baselines on module parameters is resolved in favour of the Master.

Independent Technical Reviewer appointment is required no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit is claimed. Approval authority is TBD after independent review.

— END OF DOCUMENT —

MCRS-FE-DOC-00-R7.03 · Electric Host Engineering Baseline (Corrected) · 300 kWh usable primary · T1 NDA Controlled